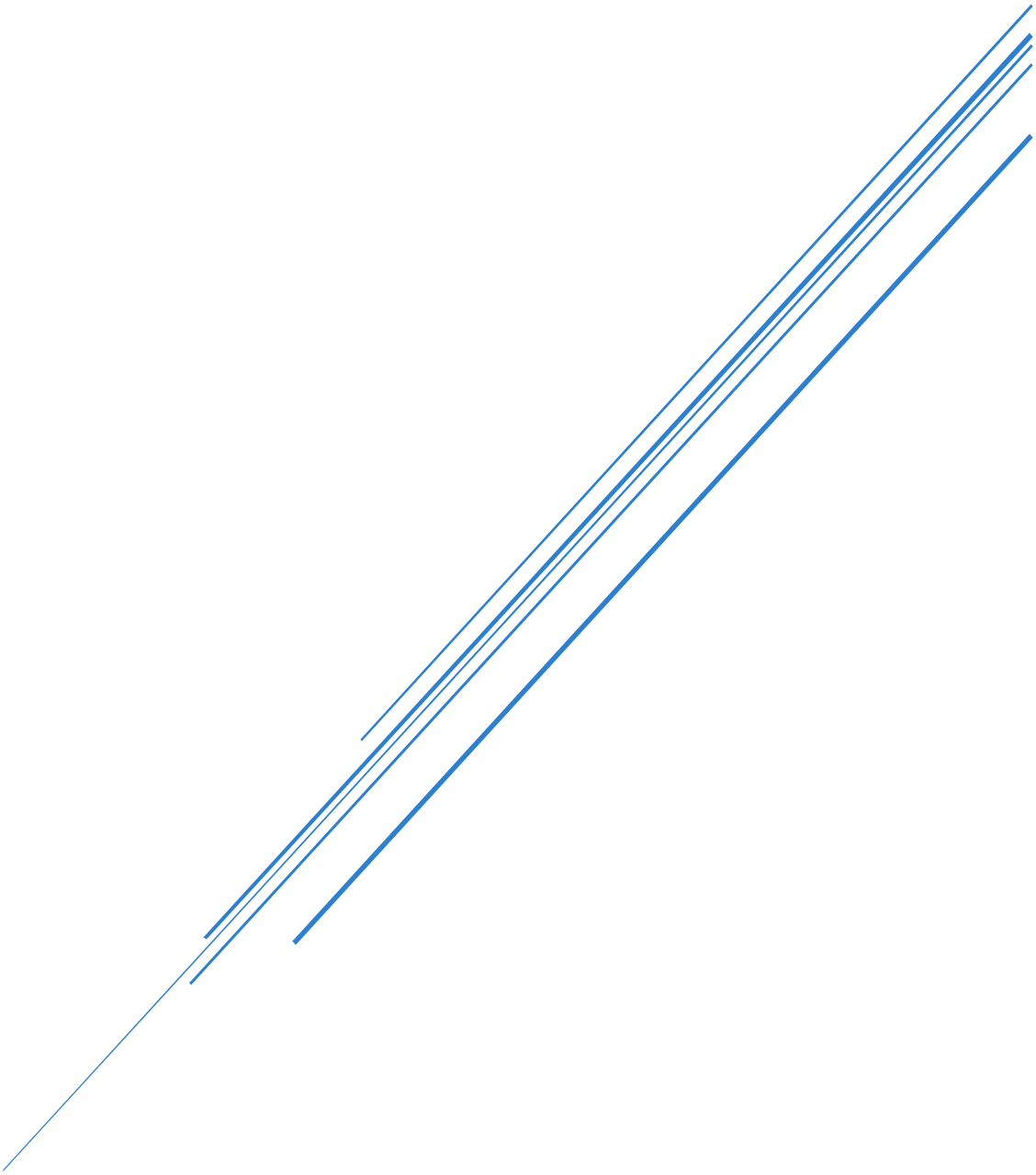
MINI PROJECT 4:

Cyber Security Awareness



By: Aiman, Seerat, Faris, Jimmy

Table of Contents

Safe use of Devices: 13

Importance of Device Update: 13

Defining the Vulnerability, Patch, and Exploit Dynamic: 13

Vulnerability:..... 13

Patch:..... 13

Exploit: 14

What roles do device encryption, screen locks, and reputable security software play in protecting data? 14

Device Encryption: 14

Screen Locks: 14

Reputable Security Software: 14

The Dangers of Unofficial Sourcing: 14

Typical Malware Delivery Pathways: 15

Hardware Port Vulnerabilities: 15

Unknown USB Devices:..... 15

Public Charging Points:..... 15

Documented Evidence for Hardware Threats: 15

USB Threats (BadUSB): 15

Public Charging Risks: 16

Safe Use of Public Wi-Fi:..... 17

Introduction 17

What specific risks arise on open public networks? 17

Eavesdropping 17

Man-in-the-Middle (MITM) Attacks 17

Evil Twin Hotspots 17

How does a VPN protect a connection, and what does HTTPS do?..... 17

HTTPS 17

VPN..... 18

HTTPS vs VPN Comparison..... 18

What practical precautions do security authorities recommend? 18

How Has Public Wi-Fi Advice Changed? 19

References 20

Introduction:

Cyber-security is now one of the most critical problems in today's digital era both for people and for organisations. The internet-connected technologies are critical for the modern society to communicate, bank money, shop, study, go to the doctor, and conduct business. These technologies offer much to the users but can also be used by cybercriminals to attack users in many ways. With technology becoming increasingly commonplace, the need for cyber-security awareness is also growing.

There are cyber-attacks that do not take the advantage of technology weaknesses, but the lack of users' understanding of the risks. Common ways that individuals are

targeted by cybercriminals include weak passwords, phishing, unsafe browsing, and insecure network connections. Organisations with good technical security can be breached if the employees do not practice good cyber-security. That's why user awareness is one of the most critical elements of cyber-security.

The five key areas for cyber-security awareness that are examined in this report are as follows: password hygiene, two or multi factor authentication, phishing and social engineering, safe use of devices, and safe use of public wi-fi. Each section explores typical threats, the risks it poses from poor security measures and what can be done, both personally and as an organisation, to address these risks.

The aim of this report is to raise awareness about the existing cyber-security risks and to promote safe online practices. Users can understand the modus operandi of cybercriminals and know what to look for in common attacks, to dramatically lower the risk of falling victim to cybercrime. The report offers guidance on how people and organisations can help make their accounts, devices, networks and sensitive information more secure in a more connected world, based on research and analysis of these five topics.

Password Hygiene:

Password hygiene is the best practice when it comes to setting up, using and protecting passwords and ensures that personal and organisational accounts are secure and safe from unauthorised access. Passwords are one of the most used tools to secure data in various areas including business files, banking accounts, university systems, emails, and many more. Poor password hygiene can leave individuals and organisations vulnerable to cyber-attacks, for example if passwords are reused across multiple accounts or simple passwords are used then it is easier for cyber criminals to gain access to sensitive information. Therefore, good password hygiene is important when managing cyber security awareness in the workplace. (National institute of standards and technology 2025).

Weak passwords and risks

A weak password is one that can be easily guessed or discovered by an attacker. Many people choose weak passwords as they are easier to remember. This however can pose security risk. (National institute of standards and technology 2025) . Some examples of weak passwords are:

Password123

123456789

Qwerty123

Or using date of birth or name

And using the same password across multiple accounts

If an attacker gains access to one account, they might try their luck with the same password on other accounts. (National institute of standards and technology 2025). For Organisations such as Northwest Architecture Studio Ltd weak passwords can allow attackers access to employee data, financial data, clients and architectural information, and cause damage to the organisation.

Common password attacks

Attackers use many techniques to gain access to passwords and unauthorised access to accounts. Some of the most common are:

Brute force attacks

Using this attack method an attacker will try many different password combinations until the correct password is found. Short and simple passwords tend to be easier to crack using this method.

Dictionary attacks

These attacks try to use lists of words, phrases and previously leaked passwords to gain access to an account. Passwords that contain common words or predictable patterns are most vulnerable to this attack.

Credential stuffing

An attacker can attempt to use credentials (usernames and passwords) stolen from one data breach to gain access to other accounts. This is especially an issue where users reuse passwords across multiple websites. (open worldwide application security project)

How to create strong passwords

Creating strong passwords is one of the most useful methods in ensuring accounts are secure from unauthorised access. The UK national Cyber Security Centre (NCSC) recommend using three random words to create a password that is longer and easier to remember. For example, a password such as TigerCloudRiver is often more secure than a password such as password 123.(National institute of standards and technology 2025)

Good password practice includes:

Unique and long passwords for important accounts At least 8 characters

Avoid using personal information in the password

Never share passwords with anyone

Each account should have a unique and different password

Multifactor authentication wherever allowed

If a password is compromised change it immediately

These above practices help reduce risk of attackers gaining access to sensitive information or guess your password.

Password managers

A password manager is a security tool which can be used to create, store and manage passwords safely. Rather than needing to remember individual passwords manually, individuals can use a password manager to access their saved passwords. These passwords are protected by a strong master password or another strong authentication method. Password managers can also create unique and long passwords for different accounts reducing the risk of password reuse, for example an employee at Northwest Architecture Studio Ltd may need separate passwords for a company system, project management software and email accounts. A password manager can assist this

employee in managing these accounts without the use of a repeated or simple password. (national cyber security centre National institute of standards and technology 2025)

Password Hygiene in Organisations

Password Hygiene is important to organisations as their employees have access to a range of sensitive business data. Northwest Architecture Studio Ltd should encourage their employees to adopt good password security practices to reduce the risk of a cyber-attack. As organisations typically have a quite moderate number of employees it is important that each employee is properly trained on password hygiene as even one employee might cause the company damage if they have a weak password, which is a door for cyber groups to attack.

Some password security recommendations for organisations include:

Regular training on cyber security awareness

Use of unique and strong passwords

Multi factor authentication

Use of secure password managers

Prevention of password sharing between employees

Monitoring of accounts for suspicious login activity

Immediate change of compromised passwords

If an employee's password were to be compromised an attacker may gain access to internal business systems, employee details, client information and architectural designs. By implementing good password security practices organisation an reduce the risk of data breaches, financial loss of reputational damage. (national cyber security centre National institute of standards and technology 2025)

Conclusion

Password hygiene is important for both individuals and organisations when it comes to cyber security. Poor password practices and password reuse can leave accounts vulnerable to dictionary attacks, credential stuffing and brute force attacks. By using unique long passwords that are strong paired with multifactor authentication and secure password managers account security can be greatly improved. Good password hygiene for Northwest architecture Studio LTD can help ensure that employee data, company data and client data and information remains secure from unauthorised

access or the company could face legal charges and loss of reputation due to malpractice in proper password etiquette.

Two Factor/Multi Factor Authentication:

Introduction

Multi-factor authentication (MFA) is a security Feature that requires users to provide two more pieces of evidence to identify themselves before they can access their account (ang. checkable and Zhou 205). Password alone is not enough because only provides only one layer of security: if an attacker gain access to the password, they can enter the account. With MFA that attacker's goal can still be stopped, because they would also need physical access to the user's phone, fingerprint or security key to access the account.

This section discusses the authentication factors used in MFA and its benefits, such as stronger security and better protection of data. It then examines the weaknesses and the challenges that MFA faces every day, including losing access to the second factor, phishing and the theft of the user's device. We will cover the usability and accessibility of MFA. At the end will we give recommendations on planning account recovery and using phishing resistants methods.

Authentication factors

MFA is based on three types of authentication factors. The first is the knowledge-factor, which represents the category of credentials that the user knows. such a password, a pin or a security questions. The second is the possession factor, which is something the users has, such as a mobile phone. In the modern security, the phone is often used as the possession factor, for example, through Authenticator apps. The third one is the inheritance factor, which is something the user has had from birth, Such as fingerprints or their face. This factor is widely used today, for example when a phone asks for fingerprint or face recognition to unlock it. And the Biometric offers convenience to the users, because this type of authentication is fast: the users only need place their fingers or show their faces when they ask to do so. But if biometrics data is compromised, the stored template (fingerprint and Face ID) can be deleted. And the user cannot change their fingerprints or Face ID in the same way as password. Factors can also be combined in one step: Marasco et al. (2023) tested finger PIN, a scheme that combines a fingerprint with a pin and found that most of the participants in the experiment did find it easy to use.

The factors of MFA must be of different types, because if they belong to the same group, they do not offer additional protection to the user. With factors from different groups, an attacker would carry out independent attacked to break each one. A common place example of MFA is online banking. Many banking apps only work on the user's registered phone, which acts as the possession factor and then ask biometric

recognition such as the user's face or fingerprint. If the biometric fails, the app asks for the pin instead.

Benefits of MFA

The main benefit of MFA is that it can block an attacker from accessing an account. Even when the password has been stolen. The first step of logging in is the password. If the password is correct, the user is taken to the second stage, where the platform asks for the second factor. If the second factor is wrong, the platform refuses access. Although it may allow another attempt. The second factor is usually a code from an authenticator app or a confirmation notification on the user's phone. In this way MFA blocks most unauthorised login attempts, because the attacker does not have the second factor. Jacomme and Kremer (2021) analysed google 2-step and FIDO security and malware and showed that the second factor can add very important protection if a password is comprised.

Another benefit is protection against password reuse. Many people use the same password for several websites and app accounts. If one pf these websites suffers a data breach, attackers may discover the password and try it one the user's other accounts, this attack is called credential stuffing. However, with MFA enable the attackers would still be missing the second factor, so they could not access the other accounts.

This is also significant in the business world, security is the foundation of a company, because companies hold sensitive customer data. Under the UK GDPR an organisation without reliable protection, can risk a heavy fines and legal consequences. However, the users still need to be careful of phishing, because a criminal can pretend to be a trusted institution to steal passwords.

Weakness of MFA

One of the biggest weaknesses of MFA is advanced phishing attacks. The most common MFA methods are codes received by SMS or codes generated by an app using TOTOP (Time-based one-time Password). If a user visits a fake website, the site can intercept not only the passwords but also the MFA codes that the user enters and the attackers can then use it immediately on the real websites. Jacomme and Kremer (2021) through their research it showed that some MFA protocols such as code-based google 2-step can still be attacked in this way when the user is phished.

Another weakness that needs to be considered is MFA fatigue. Attackers who already know the user's password can manipulate the user by sending repeated push notifications asking them to approve a login. This can cause confusion or fatigue, and the suer may end up accepting one of the notifications just to make them stop. This type of attacks is known a push bombing. In addition, users can become accustomed

to delays that an unexpected notification is simply a system errors and approve it without thinking.

Another point that needs to be discussed is SIM swapping. this is a scam in which cybercriminals trick the user's mobile network provider into transferring the user's SIM card to their control. With access to the numbered, attackers can bypass SMS-based MFA, because the one-time password (OTP) is sent to them, and they can change the password and locker the suer out (Jover,2020).

Finally, losing the phone can lock users out of their account, especially because users who rely on biometrics, such as face ID or SMS codes often forget their password. If the phone is stolen and not poorly locked there may also receive the varication codes and access the accounts.

Usability and accessibility

One of the reasons people find MFA annoying is that it takes extra tome log in, while users usually want access quickly as possible. This is linked to inconsistency, because different platforms use different methods, such as SMS, email or authenticator apps. In some cases, users also must switch between devices for example, a user logging in a on a laptop may be required to use their phone to verify the code. Research supports this a study of 400 users found that only about half used MFA, and only just over half of those were satisfied with it, which the auditors linked to designs that do not focus on the needs of users (Nanda et al.,2024).

MFA is also not equally accessible to all demographic groups. Older people for example, may find it difficult because they are less familiar with digital technologies, and being asked to install additional security methods, such as an authenticator app, can become a burden for them. Furthermore, MFA is not always inclusive, because some people with disabilities face physical or sensory limitations, and in these cases, MFA cannot offer them an optimal service.

Recommendations

Since phishing and SIM swapping are major weakness, user should be offered safer alternatives. To protect against SIM swapping, users can use an authenticator app instead of SMS, because app generates the code on the phone itself (Jover,2020). The secret key used to create the codes is stored on the device when the suer scans a QR code during setup, so the codes do not travel through the mobile network. However, an authenticator app does not fully protect against phishing, because a fake website can still steal the code. For this reason, setting up a passkey can help users to be safer. Because a passkey is linked only to the real website and cannot be used another website, which blocks the attempts of cybercriminals.

Thanks to recent innovations, the waves of notifications sent by attackers can also be stopped through number matching. With number matching, the login screen shows a

random number, and the suer must enter that number in the app on their phone, instead of simply choosing “approve” or “deny” this is reliable because the number is generated at the moment of login and is only shown on the screen of the person trying to log in, so a user who is not logging in cannot approve the request by mistake.

In case users lose their phone, it is very important that they prepare three things: multi-device synchronisation, an alternative email address and backup codes. Setting these up is essential, because otherwise users' risk being permanently locked out of their accounts and losing important digital data, such as sensitive documents stored on the phone and conversation histories linked to the cloud.

Conclusion

In conclusion, MFA is useful because it provides multiple layers of protection for the user. However, MFA also has well-known limitations, such as MFA fatigue (push bombing), SIM swapping and phishing, which are problems that all users, as well as both large and small organisation, are exposed to. To this shows that reliable and trustworthy security does not come only from adding more steps for the user, but from choosing verifications methods that cannot be easily compromised, such as authenticator apps, number matching and passkeys. For this reason, user and business should educate themselves as much as possible and keep up with new methos that attackers develop.

Phishing and Social Engineering:

Phishing is one of the most prevalent types of cybercrime and is a major danger to both individuals and organisations. Phishing attacks are based on human psychology and trust rather than technical weaknesses, as is the case with many types of cyber-attacks. Attackers try to trick their victims into giving out confidential information or downloading harmful software and/or commands which are advantageous to the attacker. Phishing is not a technical hacking skill, so it remains a very effective attack, even when organisations have high technical security measures. (National Cyber Security Centre, n.d.).

The more general term for these types of psychological manipulation methods is social engineering. Phishing is one of the most prevalent forms of social engineering due to its ability to leverage on emotions like urgency, fear, trust and curiosity, to convince users to undertake actions without taking a second thought to the implications. (Cybersecurity and Infrastructure Security Agency, 2021).

What is Phishing?

Phishing is a form of cyber-attack that involves fraudulent messages from attackers posing as trusted organizations, people or services to obtain confidential information. Such information can contain usernames, passwords, banking information, personal and company information.

Emails, SMS messages, phone calls or fake websites that look like the real ones are frequently used to attack. Its aim is generally to obtain credentials or install malware, gain access to business networks, or engage in financial fraud.

One of the most serious things about phishing is that they can come across as legitimate. Often, the branding, logos, language, formatting and other characteristics of legitimate organisations, like banks, universities, online retailers and government agencies, are replicated, so that it can be hard to tell the difference between legitimate and fake communications. (National Cyber Security Centre, n.d.)
There are three categories of phishing attacks:

Traditional Phishing:

The classic phishing scam is an identical fraud e-mail sent to a mass of recipients. The attacker hopes that a few of the users will click on the message and give away their sensitive information. (NCSC, n.d.).

Spear Phishing:

Spear phishing is a more targeted type of phishing that targets individuals and/or organisations. An attacker will normally do some research on a victim before sending messages to them.

An attacker can, for instance, send an e-mail claiming to be a manager and ask for payroll information from a member of the Human Resources department. (National Cyber Security Centre, n.d.)

Smishing:

Smishing is a phishing attack via SMS text messages. These messages frequently include links that look suspicious, and the messages say there is a need to take immediate action.

For instance, if someone receives a text message that inform them that their parcel delivery has been missed and to click a link to reschedule the package. (Cybersecurity and Infrastructure Security Agency, 2021).

Vishing:

A form of phishing that takes place over the telephone, in which attackers pretend to be trusted organisations like banks, internet providers or government agencies.

The attacker tries to convince the victim to give out personal details or send money.

Business Email Compromise (BEC):

BEC is an advanced phishing scam where the scammer pretends to be a high-ranking executive, supplier or trusted partner.

Often these attempts are directed at a business and try to encourage anyone on their staff to make a false payment, or to release confidential business information. The BEC attacks have resulted in substantial losses to companies all over the world since workers might think that the request is from a trusted source.

The reasons **why phishing attacks are effective.** The rationale behind the success rate of phishing attacks.

Attackers are not only successful because of phishing exploits, but also because they exploit human psychology. Cybercriminals know people well and know how they will behave when stressed. (Cybersecurity and Infrastructure Security Agency, 2021)

Urgency:

False urgency: attackers try to make the victim feel as though it's an emergency.

Examples include:

- You will be informed that your account will be closed today.

- The payment must be made right away.
- The account that you used to attend the university has been suspended.

If the user feels compelled, he or she is not likely to be thorough in the assessment of the message.

Authority:

Most attackers pretend to be authority badges, e.g. managers, banks, universities or government organisations.

Individuals are more inclined to obey requests from people that they feel are in authority.

Fear:

Another psychological technique is fear. Malicious actors can threaten to take legal action if immediate action is not taken or that the recipient account has been compromised.

These messages are meant to invoke emotions and not logic.

Curiosity:

Some phishing messages try to take curiosity by offering intriguing chances, gifts, or vital records.

Examples include:

- You've won a prize!
- You may now see your confidential report.
- "Important examination results."

A common mistake is users clicking through a link without realizing it's a scam. Users are often tempted to click on links before realizing that it is a scam. (Cybersecurity and Infrastructure Security Agency, 2021).

Trust:

Often attackers pretend to be well-known brands and organisations. Messages that look as if they're from a company that the recipient regularly communicates with are more likely to be trusted.

The following are some of the usual indicators of a phishing message.

The user should be aware of some warning signs that could mean the communication is fraudulent. (National Cyber Security Centre, n.d.)
Common indicators include:

- Unexpected e-mails or messages.
- Use of a generic salutation e.g. 'Dear Customer'.
- Inadequate grammar, spelling or formatting.
- Asks for passwords or financial information.
- Suspicious website links.
- Unexpected attachments.
- No messages are required to take urgent action.
- Email addresses of sender that are different from the claimed organisation.
- Requests for payment that are out of the ordinary for business transactions.

Many phishing messages don't include all the warning signs listed here, but having a few of them present at the same time should be a red flag for the user to take further action before responding to a message.

What to do when you **encounter a Phishing Attempt:**

When a user suspects a phishing attempt, he or she should take immediate action.

Recommended steps include:

- Avoid opening links and clicking on attachments.
- Check the message with official channels of communication.
- Inform the organisation's IT or security department of the incident.
- To remove the message after it is reported.
- If the logon was successful, change the password as soon as possible.
- Use 2-factor authentication where possible.
- Keep an eye on for suspicious activities.

If it is suspected that malware may have been downloaded, the device should be isolated and security staff should get to work.

Phishing poses a threat to organisations in the following ways: For organisations, phishing is a threat:

Organisations can suffer from bad consequences because of phishing. Successful attack could lead to:

- Data breaches.
- Financial losses.
- Identity theft.
- Ransomware infections.
- Reputational damage.
- Interruption of business activities.
- Loss of confidential client information.

If an attack on a firm like Northwest Architecture Studio Ltd were to be successful, it could reveal sensitive project files, architectural designs, financial records and confidential client data. This may result in a legal liability, loss and damage to the trust of the client. (National Cyber Security Centre, n.d.)

Preventing Phishing Attacks:

The best way to minimize the threat from phishing is to have a combination of technical controls and staff awareness training.

Security measures include:

- Frequent Cyber Security Awareness Training.
- Multi-factor authentication.
- Email filtering systems.
- Strong password policies.
- Ensuring software updates and patch management.
- Verification procedures for financial transactions.
- Incident reporting processes.

Technical controls are a very important line of defence, but employees are also a line of defence, and many phishing attacks are successful when an employee unwittingly helps the attacker. (National Cyber Security Centre, n.d.)

Conclusion:

Phishing and social engineering remain two of the biggest cyber-security risks for people and organisations today. Phishing attacks are different from technical exploits in that they rely on the gullibility, urgency, fear and curiosity of human users. The diversity of phishing methods, such as spear phishing, smishing, vishing and BEC, shows the versatility of the phishing with the intent to create as much success as possible. Industry best practices and knowledge of phishing indicators can help users minimize their risk of being phished. Being vigilant and keeping up to date with awareness training alongside robust organisational security measures is one of the best methods of preventing phishing attacks and securing sensitive data.

Safe use of Devices:

Device protection represents the critical final line of defence within modern digital environments. Because network-level security controls can only filter traffic up to a certain point, while individual physical assets such as laptops, smartphones, and tablets serve as the final repositories for sensitive personal information, corporate accounts, and private financial records. Cybercriminals increasingly target these physical endpoints directly, exploiting unpatched systems, weak user configurations, or malicious hardware accessories to gain unauthorized entry. (National Institute of Standards and Technology,2023)

The following technical breakdown examines how software update lifecycles, local storage encryption, software sourcing choices, and hardware port access methods function to improve the protection against exploitation of laptops, phones and other digital devices.

Importance of Device Update:

According to official documentation from the UK National Cyber Security Centre (NCSC), modern mobile and desktop devices run an immense volume of software ranging from the base operating system to complex local utilities like office applications, web browsers, and antimalware packages. While hardware manufacturers regularly issue updates to introduce fresh functionality or general system performance improvements, installing these updates promptly is a core security requirement (National Cyber Security Centre, 2025).

The NCSC notes that software updates frequently contain security patches designed to address vulnerabilities before they can be exploited by attackers. If updates are not installed, cybercriminals may identify outdated systems and exploit known weaknesses to steal information, deploy ransomware, or disrupt device operations (National Cyber Security Centre, 2026).

Defining the Vulnerability, Patch, and Exploit Dynamic:

The protective interaction between system developers and external adversaries operates within a strict technical lifecycle based on the following three elements:

Vulnerability:

A vulnerability is a weakness, flaw, or misconfiguration within software or hardware that could allow an attacker to gain unauthorised access, manipulate information, or disrupt normal operations (National Cyber Security Centre, 2026).

Patch:

A patch is a software update released by a vendor to correct identified vulnerabilities and improve system security. Applying patches promptly reduces the opportunity for attackers to exploit known weaknesses (National Cyber Security Centre, 2026).

Exploit:

An exploit is a tool, program, or method used by attackers to take advantage of a vulnerability and perform unauthorised actions on a system before security controls are implemented (National Cyber Security Centre, 2026).

What roles do device encryption, screen locks, and reputable security software play in protecting data?

Protecting data stored locally on a device requires a defence-in-depth model that combines physical and cryptographic barriers to ensure data cannot be extracted if a machine is physically lost, misplaced, or stolen:

Device Encryption:

Full-Disk Encryption (FDE) platforms, such as Microsoft BitLocker or Apple File Vault, secure data at rest by converting information into unreadable ciphertext. If a laptop or smartphone is lost or stolen, encryption prevents unauthorised individuals from accessing stored files without the correct decryption key (National Institute of Standards and Technology, 2025).

Screen Locks:

Screen locks serve as the primary physical access barrier for device protection. Forcing complex passcodes or biometrics (such as capacitive fingerprint scanning or infrared facial mapping) prevents immediate access. This role is critical because if an unauthorised user gains access to an active, unlocked device, they completely bypass full-disk encryption and local security permissions, gaining full access to private data stores (National Institute of Standards and Technology, 2023).

Reputable Security Software:

Reputable security software provides continuous monitoring for malicious activity. Modern anti-malware solutions use behavioural analysis, cloud-based intelligence, and threat detection techniques to identify and block malware before significant damage occurs (National Institute of Standards and Technology, 2023).

The Dangers of Unofficial Sourcing:

Installing software from unofficial websites, peer-to-peer networks, or third-party repositories significantly increases security risks. Unlike official application stores, these sources may not perform security checks before distributing software, increasing the possibility of hidden malware, spyware, keyloggers, or ransomware being bundled with legitimate applications (Cybersecurity and Infrastructure Security Agency, 2025)

Typical Malware Delivery Pathways:

Malware traditionally reaches a device through user-driven installation vectors:

Phishing Attachments: Opening malicious email attachments containing malware or harmful macros remains one of the most common methods used to infect devices (Cybersecurity and Infrastructure Security Agency, 2025).

Deceptive Malvertising Banners: Clicking on fake update prompts or fraudulent advertisement can trick users into downloading malicious software (Cybersecurity and Infrastructure Security Agency, 2025).

Drive-By Downloads: Visiting a compromised website that uses browser exploit kits to drop malware onto unprotected devices without the user's active consent (Cybersecurity and Infrastructure Security Agency, 2025).

Hardware Port Vulnerabilities:

Unknown USB devices and public charging terminals introduce profound physical layer hardware vulnerabilities to an individual device:

Unknown USB Devices:

Unknown USB devices can present significant security risks because attackers may modify them to impersonate legitimate input devices such as keyboards. Once connected, these devices can automatically execute commands and install malicious software without the user's knowledge (Cybersecurity and Infrastructure Security Agency, 2024)

Public Charging Points:

Public charging stations may expose users to risks commonly known as "juice jacking". Because USB connections support both power transfer and data communication, compromised charging stations could potentially attempt to access device information or install malware. While confirmed real-world attacks remain limited, security organisations continue to recommend using personal chargers, power banks, or charge-only cables when charging devices in public locations (Roppolo, 2023; Cybersecurity and Infrastructure Security Agency, 2024).

Documented Evidence for Hardware Threats:

USB Threats (BadUSB):

A well-documented example of USB-based attacks is the BadUSB research first presented by security researchers at the Black Hat conference in 2014. The researchers demonstrated how the firmware of a USB device could be reprogrammed to impersonate trusted peripherals such as keyboards, allowing malicious commands to

be executed automatically when connected to a computer. The findings highlighted that traditional antivirus software may struggle to detect such attacks because the operating system interprets the device as legitimate hardware rather than malicious software. This research provided evidence that USB devices can be used as a delivery mechanism for cyberattacks and remains a significant reference point in USB security discussions (Nohl and Lell, 2014).

Public Charging Risks:

In April 2023, the Federal Bureau of Investigation (FBI) Denver Field Office issued a public warning advising users to avoid free charging stations located in airports, hotels and shopping centres. According to the FBI, malicious actors may use compromised USB charging ports to introduce malware or monitoring software onto connected devices. Although the warning was precautionary and not linked to a specific incident, it demonstrates that public charging infrastructure is considered a credible security risk by law-enforcement agencies (Roppolo, 2023).

Safe Use of Public Wi-Fi:

Introduction

Public Wi-Fi networks have become common in locations such as airports, hotels, cafés, libraries, universities and shopping centres. While these networks provide convenient internet access, they can also expose users to cyber-security risks if used incorrectly. Attackers may exploit unsecured wireless networks to intercept communications, steal personal information or impersonate legitimate networks. As a result, understanding the risks associated with public Wi-Fi and the measures available to reduce these risks is an important aspect of cyber-security awareness (National Cyber Security Centre, 2025).

What specific risks arise on open public networks?

Eavesdropping

Eavesdropping occurs when an attacker intercepts network traffic transmitted across a public network. If information is transmitted without adequate encryption, attackers may be able to view sensitive data such as usernames, passwords, emails or financial information. Encryption helps prevent this by making intercepted information unreadable to unauthorised parties (National Cyber Security Centre, 2025).

Man-in-the-Middle (MITM) Attacks

A Man-in-the-Middle (MITM) attack occurs when an attacker secretly positions themselves between a user and the service being accessed. This allows the attacker to monitor, intercept or modify communications without either party being aware. Public

Wi-Fi environments can increase the opportunity for these attacks because multiple users share the same network infrastructure (National Cyber Security Centre, 2025).

Evil Twin Hotspots

An Evil Twin attack occurs when a cybercriminal creates a fake wireless access point that appears identical or like a legitimate public Wi-Fi network. Users may unknowingly connect to the malicious network believing it to be genuine. Once connected, attackers may monitor traffic, capture credentials or redirect users to fraudulent login pages. The Cybersecurity and Infrastructure Security Agency identifies Evil Twin networks as a recognised attack technique used to deceive users into connecting to malicious Wi-Fi hotspots (Cybersecurity and Infrastructure Security Agency, 2025).

How does a VPN protect a connection, and what does HTTPS do?

HTTPS

Hypertext Transfer Protocol Secure (HTTPS) encrypts communications between a user's web browser and a website. This protects information from being read or altered during transmission. HTTPS also helps verify that users are communicating with the legitimate website rather than an impersonator. The Information Commissioner's Office states that HTTPS protects website communications through encryption and reduces the risk of interception by third parties (Information Commissioner's Office, 2025).

VPN

A Virtual Private Network (VPN) creates an encrypted tunnel between a device and a VPN server. According to the National Cyber Security Centre, VPNs are encrypted network connections that help protect data in transit across untrusted networks. VPNs provide an additional layer of protection when using public Wi-Fi networks because attackers cannot easily view the encrypted traffic travelling through the VPN tunnel (National Cyber Security Centre, 2025).

HTTPS vs VPN Comparison

Feature	HTTPS	VPN
Encrypts website traffic	Yes	Yes
Protects all internet traffic from device	No	Yes
Verifies website authenticity	Yes	No
Protects data on public Wi-Fi	Partially	More comprehensively
Works automatically on modern websites	Usually	Requires setup

What practical precautions do security authorities recommend?

To reduce risks when using public Wi-Fi, security authorities recommend:

- Only connect to trusted and verified Wi-Fi networks.
- Confirm the official network name before connecting.
- Avoid accessing highly sensitive accounts on unknown networks.
- Keep devices and software updated.
- Disable automatic Wi-Fi connections.
- Ensure websites use HTTPS.
- Consider using a VPN when transmitting sensitive information.
- Disconnect from public Wi-Fi when it is no longer needed.
- Avoid sharing files or enabling network discovery on public networks.

The National Cyber Security Centre notes that attackers on insecure networks may attempt to intercept or modify communications, making secure connections and encryption important safeguards. (National Cyber Security Centre, 2025)

How Has Public Wi-Fi Advice Changed?

Older cyber-security guidance often advised users to avoid public Wi-Fi whenever possible because network traffic was frequently transmitted without encryption. However, modern internet services increasingly use HTTPS encryption by default. As a result, current guidance focuses less on completely avoiding public Wi-Fi and more on ensuring that secure protocols, software updates, and encryption are being used correctly.

Government guidance has also evolved. Earlier wireless-network recommendations often required VPN usage for wireless access, whereas more recent guidance acknowledges that widespread encryption and improved security technologies have changed the risk landscape. (National Cyber Security Centre, 2025; Information Commissioner's Office, 2025).

Therefore, while public Wi-Fi can still present security risks, users who follow current security recommendations and make use of encryption technologies can significantly reduce their exposure to cyber threats.

Conclusion:

Being aware of cyber security is a key aspect of life in the modern digital age. Technology advances and internet-connected services become more of the norm and cybercriminals are continuously evolving their methods to attack individuals and organisations. Phishing attacks, credential theft, malware and network-based attacks can have devastating effects, such as financial loss, data breaches, identity theft and damage to reputation.

Throughout this report, the five themes mentioned emphasize the need to be safe and responsible online. Even if user credentials are compromised, multi-factor authentication will add another layer of security, and strong password hygiene will help prevent unauthorized access to user accounts. Having knowledge of phishing and social engineering methods can help users identify and prevent being tricked into providing personal information. In the same way, knowing how to securely and safely connect devices to public networks can minimize their vulnerability to malware, unauthorised entry and data interception.

One major message in this report is that cyber-security is not a technical task or a job for the security team. All of those who use digital technologies have a vital role to play in security. Educational awareness and knowledge are key weapons in the fight against cybercrime, as many successful attacks occur due to human error.

In conclusion, establishing a healthy cyber-security routine can help minimize risk and create a safer online environment for all. The recommendations outlined in this report will help individuals and organisations to be better prepared to detect threats, secure sensitive data and cover for potential security incidents. In today's inter-connected world, where cyber threats are continually changing, continuous learning and awareness are ever more vital.

References

Cybersecurity and Infrastructure Security Agency (CISA) (2021) *Avoiding Social Engineering and Phishing Attacks*. Available at: https://www.cisa.gov/news-events/news/avoiding-social-engineering-and-phishing-attacks (Accessed: 14 October 2026).

National Cyber Security Centre (NCSC) (n.d.) *Phishing attacks: defending your organisation*. Available at: https://www.ncsc.gov.uk/guidance/phishing (Accessed: 14 October 2026).

National Cyber Security Centre (NCSC) (n.d.) *Exercise in a Box: Phishing*. Available at: https://www.ncsc.gov.uk/information/exercise-in-a-box (Accessed: 14 October 2026).

Microsoft (n.d.) *Protect yourself from phishing*. Available at: https://support.microsoft.com (Accessed: 14 October 2026).

IBM (n.d.) *What is Phishing?* Available at: https://www.ibm.com/topics/phishing (Accessed: 14 October 2026).

Norton (n.d.) *What is Phishing and How to Prevent It?* Available at: https://us.norton.com/blog/emerging-threats/phishing (Accessed: 14 October 2026).

Cybersecurity and Infrastructure Security Agency, 2024. *Secure Our World*. [Online]
Available at: https://www.cisa.gov/secure-our-world
[Accessed 15 October 2026].

Cybersecurity and Infrastructure Security Agency, 2025. *Cybersecurity Best Practices*. [Online]
Available at: https://www.cisa.gov/topics/cybersecurity-best-practices
[Accessed 15 October 2026].

Cybersecurity and Infrastructure Security Agency, 2025. *Evil Twin (T1557.004)*. [Online]
Available at: https://www.cisa.gov/eviction-strategies-tool/info-attack/T1557.004
[Accessed 15 October 2026].

Information Commissioner's Office, 2025. *Encryption and Data Transfer*. [Online]
Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/security/encryption/encryption-and-data-transfer/
[Accessed 15 October 2026].

National Cyber Security Centre, 2025. *Connecting Securely*. [Online]
Available at: https://www.ncsc.gov.uk/section/exercise-in-a-box/connecting-securely
[Accessed 15 October 2026].

National Cyber Security Centre, 2025. *NCSC Device Security Guidance Collection*. [Online]
Available at: https://www.ncsc.gov.uk/collection/device-security-guidance/managing-deployed-devices/keeping-devices-and-software-up-to-date

National Cyber Security Centre, 2025. *Virtual Private Networks (VPNs)*. [Online]
Available at: https://www.ncsc.gov.uk/collection/device-security-guidance/infrastructure/virtual-private-networks
[Accessed 15 October 2026].

National Cyber Security Centre, 2026. *Vulnerability Management*. [Online]
Available at: https://www.ncsc.gov.uk/collection/vulnerability-management
[Accessed 15 October 2026].

National Institute of Standards and Technology, 2023. *Guidelines for Managing the Security of Mobile Devices in the Enterprise (SP 800-124 Rev. 2)*. [Online]
Available at: https://csrc.nist.gov/pubs/sp/800/124/r2/final
[Accessed 15 October 2026].

National Institute of Standards and Technology, 2025. *Securing Data & Devices*. [Online]
Available at: https://www.nist.gov/itl/smallbusinesscyber/guidance-topic/securing-data-devices
[Accessed 15 October 2026].

Nohl, K. & Lell, J., 2014. *BadUSB: On Accessories That Turn Evil*. s.l., s.n.

Roppolo, M., 2023. FBI office warns against using public phone charging stations at airports or malls, citing malware risk. CBS News., 2023. *Public Charging Station Security Warning*. [Online]
Available at: https://www.cbsnews.com/news/fbi-warns-against-juice-jacking-what-is-it/
[Accessed 15 October 2026].

National Cyber Security Centre (NCSC) (n.d.-b) Three random words. Available at: https://www.ncsc.gov.uk/collection/top-tips-for-staying-secure-online/three-random-words (Accessed: 16 October 2026).

OWASP Foundation (n.d.) Credential Stuffing Prevention Cheat Sheet. Available at: https://cheatsheetseries.owasp.org/cheatsheets/Credential_Stuffing_Prevention_Cheat_Sheet.html (Accessed: 16 October 2026).

National Cyber Security Centre (NCSC) (n.d.-a) Password managers: using browsers and apps to safely store your passwords. Available at: https://www.ncsc.gov.uk/collection/top-tips-for-staying-secure-online/password-managers (Accessed: 16 October 2026).

Ang, K.W.,Chekole, E.G and Zhou, J. (2025) 'Unveiling the covert vulnerabilities in multi-factor authentication protocols: A systematic review and security analysis; ACM Computing Surveys,57(11), pp.1-37.doi:10.1145/3734864.

Jacomme, C. and Kremer, S. (2021)' An extensive formal analysis of multi-factor authentication protocols', ACM Transactions on privacy and security, 24(2). Article 13, pp. 1-34.doi:10.1145/3440712

Jover,R.P.(2020)'security analysis of SMS as a second factor of authentication',ACM queue,18(4), pp.37-60

Marasco,E.albanese,M., Patibandla V.V.R, Vurity, A. and sriram,S.S. (2033) 'Biometric multi-factor authentication: On the usability of the FingerPIN scheme', Security and Privacy,6(1), e261.doi.10.1002/spy2.261

Nanda, A.,Jeong,J., Shah.S.W.A., Nosouhi,M.and Doss, R.R.M. (2024)'Examining usable security features and user perceptions of Physical Authentication Devices', Computers & Security,139, Article 103664.doi: 10.1016/j.cose.2023.103664